

My-Finance-Tracker

Implemented Features

Generated July 21, 2026

AI-ready Personal Finance, Investment & Loan Analytics Platform

Stack: FastAPI + SQLAlchemy + SQLite (backend) | React + TypeScript + Vite + MUI + Redux Toolkit +
Framer Motion + ECharts (frontend)

1. Overview

This document lists every feature currently implemented in My-Finance-Tracker — a personal finance, budgeting, investment, loan, and trading tracker with journaling, gamification, and data-portability built in. It supersedes the July 20 edition: this build adds full loan tracking, a trading journal, net worth history, a financial health score, gamification streaks, tagging, a calendar view, a SIP calculator, global search, an audit-log activity feed, and JSON backup/restore, on top of the original expense/income/portfolio/budget/goal/notification foundation.

Backend	FastAPI, SQLAlchemy, SQLite (dev), JWT auth, pytest (73 tests passing)
Frontend	React 18, TypeScript, Vite, MUI v5, Redux Toolkit, Framer Motion, ECharts
Run	Single command: <code>./run.sh</code> (see <code>RUNNING.md</code>)
Status	Fully implemented, tested, and verified end-to-end

2. Authentication & Account

Auth

- Registration with email, password, and full name
- Password hashing with bcrypt (via passlib)
- JWT access tokens + refresh tokens, with automatic refresh-on-401 in the frontend
- Login / logout
- Default expense and income categories auto-seeded for every new user

Profile & Settings

- Update full name
- Change password (verifies current password before accepting a new one)
- Dedicated Settings page with tabbed sections: Profile, Appearance, Data

3. Expense Management

- Full CRUD: create, list, update, delete expenses
- Categories with Need / Want classification, seeded with common defaults (Rent, Food, Groceries, Electricity, ... / Movies, Shopping, Travel, ...) plus custom category creation
- Archive / restore workflow (soft delete) in addition to permanent delete, with a dedicated Active/Archived toggle
- Bulk actions: multi-select rows for bulk archive, bulk restore, and bulk delete
- CSV bulk import with per-row validation, reporting how many rows were imported vs. skipped and why
- Voice input: "Add 250 for groceries"-style speech-to-text entry via the Web Speech API
- Recurring expenses: mark an expense as monthly-recurring; missed months are automatically backfilled (idempotently) whenever the list is fetched
- Tagging: attach any number of colored tags to an expense for cross-cutting labels
- Link an expense to a savings goal as a contribution, visible on both the expense row and the goal
- Date-range and category filters (server-side query params)
- Client-side search across description and category name
- Sortable table (date, category, amount) with pagination
- Pie chart and sunburst chart of spend by category, reflecting the current filtered view
- CSV export of the currently filtered/searched list
- Edit dialog, confirm-before-delete, and toast feedback on every action

4. Income Management

Mirrors the Expense module with an equivalent feature set:

- Full CRUD with categories (Salary, Freelancing, Dividends, Interest, Other, seeded by default; custom categories supported)
- Archive / restore workflow plus permanent delete, with bulk archive / restore / delete actions
- Recurring income with automatic monthly backfill
- Tagging and goal-linking, matching the Expense module
- Date-range and category filters, search, sortable + paginated table
- Pie chart of income by source
- CSV export, edit dialog, confirm-delete, toast feedback

5. Portfolio Management

- Track holdings across stocks, ETFs, mutual funds, gold, silver, FD, PPF, NPS, crypto, and bonds
- Each holding records symbol, name, asset type, quantity, buy price/date, current price, broker, sector, exchange, target price, stop loss, notes, and tags
- Computed per holding: invested value, current value, profit/loss, profit/loss %, and CAGR (annualized return from buy date to today)
- Full CRUD with an edit dialog, confirm-before-delete, and archive/restore
- Search across symbol / name / sector
- Sortable table (symbol, P&L, CAGR, current value) with pagination
- Allocation pie charts by asset type and by sector
- CSV export

Note: `current_price` is user-editable in this phase since there is no live market data feed yet.

6. Loans

- Track loans by type: home, car, education, personal, other
- Each loan records name, type, principal, interest rate, EMI amount, outstanding balance, start date, tenure (months), and notes
- Computed per loan: amount paid off, payoff progress %, and the next EMI due date (derived from start date and tenure, rolled forward to the next unpaid month)
- Full CRUD, plus archive/restore for paid-off or discontinued loans with an Active/Archived toggle
- Total outstanding balance and total monthly EMI stat cards
- Progress bar per loan card showing payoff percentage
- Feeds into the Net Worth calculation (outstanding balance reduces net worth) and the Financial Health Score (debt-to-income component)
- EMI due dates surface on the Calendar view

7. Budget

- Set a monthly spending limit per expense category (one budget per category)
- Live tracking of amount spent this month, remaining budget, and percent used, computed from actual expense records
- Color-coded progress bar (green under 80%, amber 80-100%, red over 100%)
- Full CRUD: create, edit limit, delete
- Feeds the Notifications system (80%/100% alerts) and Gamification (Budget Champion badge)

8. Goals

- Track savings goals with a name, target amount, current amount, target date, and notes
- Computed progress percentage and monthly contribution needed to reach the goal by its target date
- Full CRUD with progress bars on each goal card, plus archive/restore
- Transaction drill-down: view every expense and income entry linked to a goal as a contribution
- Goal due dates surface on the Calendar view

9. Trading Journal

- Log trades with symbol, direction (long/short), quantity, entry price/date, and optional exit price/date to close a position
- Qualitative fields per trade: strategy, emotion, mistake, and lessons learned
- Automatic profit/loss calculation on close, direction-aware (long vs. short)
- Holding period (days) computed from entry to exit date
- Analytics panel: total trades, closed trades, win rate %, total P&L, average P&L, best trade, worst trade, and average holding period
- Full CRUD with edit/delete and confirm-before-delete

10. Net Worth & Financial Health Score

Net Worth

- Net worth computed live as total portfolio value minus total outstanding loan balance
- Daily snapshot recorded automatically (upserted once per day) whenever the timeline is viewed
- 180-day net worth history chart on the Dashboard

Financial Health Score

- A single 0-100 score blending five weighted components, each rated Weak / Average / Good / Excellent:
- Savings ratio — share of this month's income not spent
- Debt — outstanding loan balance relative to annualized income
- Investment diversification — breadth of distinct asset types held
- Expense control — how well active budgets are being kept to
- Emergency fund — portfolio value as a multiple of monthly expenses (target: 6 months)
- Displayed as a dedicated Health Score card on the Dashboard with a per-component progress breakdown

11. Gamification

- No-spend streak counter: consecutive days with zero recorded spend, looking back up to 90 days
- Budget Champion badge, awarded when every active budget is at or under its limit for the month
- Badges surfaced as chips on the Dashboard, color-coded by status

12. Notifications

- Automatic alerts when a budget crosses 80% used (warning) or 100% used (error)
- Deduplicated per budget/month so the same alert is not repeated
- Notification bell in the navbar with an unread-count badge, polling every 60s
- Mark a single notification read, or mark all as read

13. Dashboard & Analytics

- Time-of-day greeting personalized with the user's name
- KPI cards: Net Worth, Today's Profit/Loss, Monthly Income, Monthly Expenses (animated count-up on load)
- 180-day net worth history line chart
- 6-month cash flow line chart (income minus expenses per month)
- 6-month expense vs. income trend bar chart
- Expense allocation pie chart (current month, by category)
- Portfolio allocation pie chart (by asset type)
- Top gainers / top losers tables from current portfolio holdings
- Financial Health Score card and Gamification badges
- All figures computed live from actual expense, income, portfolio, and loan records

14. Tags

- User-defined, colored tags that can be attached to expenses, income entries, and portfolio holdings
- Create and delete tags, with duplicate-name protection (case-insensitive)
- Tags round-trip through backup export/import, re-created or matched by name on restore

15. Calendar

- Month-grid view combining expenses, income, goal due dates, and loan EMI due dates on a single calendar
- Color-coded event chips by type (expense / income / goal / loan), with a "+N more" overflow indicator on busy days
- Month navigation (previous/next) with today highlighted

16. SIP Calculator

- Projects the future value of a monthly Systematic Investment Plan from a monthly amount, expected annual return, and duration in years
- Interactive sliders plus direct numeric entry for all three inputs
- Stat cards for total invested, estimated returns, and future value
- Year-by-year invested-vs-value bar chart

17. Global Search & Command Palette

- Global command palette (Ctrl/Cmd+K) for instant navigation to any page in the app
- Live search across expenses, income, portfolio holdings, and goals by name/description, surfaced directly in the command palette with a link to the matching page
- Quick Actions FAB for one-click "add new" shortcuts that deep-link into the right page's create dialog

18. Activity Log (Audit Trail)

- Every create, update, archive, restore, and permanent-delete action across expenses, income, portfolio holdings, loans, and goals is recorded with a human-readable summary and timestamp
- Dedicated Activity page with a chronological feed, icon and color coded by action type, and relative timestamps ("3m ago", "2d ago", ...)

19. Backup & Restore

- One-click JSON export of the full account: tags, expenses, income, portfolio holdings, loans, goals, and budgets
- Import/restore from a previously exported JSON file, matching or re-creating categories and tags by name and reporting a per-entity count of records imported
- Accessible from the Settings page's Data tab

20. System Health

- Live application health check: database connectivity, API version, environment, and on-disk database size
- Dedicated System Health page with an at-a-glance operational status banner

21. Design & User Experience

Visual design

- Custom Apple-inspired theme: Inter typeface, refined color palette, rounded cards, pill-shaped buttons
- Full light and dark mode support, including chart color theming, with a toggle in the navbar
- Four selectable accent colors (Blue, Emerald, Purple, Midnight)
- Adjustable font scale and a high-contrast mode for accessibility
- Frosted-glass navbar and sidebar (backdrop blur)

Animation (Framer Motion)

- Split-screen animated login/register hero panel with drifting gradient orbs
- Staggered field entrance animation on login/register forms
- Animated sliding active-page indicator in the sidebar
- Staggered card/chart entrance animations on the Dashboard and other pages
- Animated count-up numbers on KPI cards
- Cross-fade page transitions on route change

Usability infrastructure

- Toast/snackbar notifications for every create/update/delete action, success or failure, including undo-able archive actions
- Confirm-before-delete dialogs everywhere data can be permanently deleted
- Loading states on all save buttons
- Skeleton loaders (replacing spinners) across all major pages
- Responsive, collapsible sidebar with a hamburger menu on mobile widths
- Global command palette (Ctrl/Cmd+K) for fast navigation and search
- Custom 404 page and empty-state illustrations
- Per-page browser tab titles
- CSV export on Expenses, Income, and Portfolio; CSV bulk import on Expenses

22. Developer & Operational Experience

- Single-command startup: `./run.sh` sets up the backend venv, installs frontend deps, and launches both servers together, refusing to start if ports are already in use
- 73 backend pytest tests covering auth, expenses, income, portfolio, loans, dashboard, budgets, goals, notifications, recurring transactions, trading journal, backup/restore, audit log, and profile/password changes, all run against the real FastAPI app via TestClient
- Frontend type-checked end-to-end with TypeScript (`tsc -b`)
- RUNNING.md with full setup, troubleshooting, and Node-install instructions
- Layered backend architecture: `api` -> `services` -> `repositories` -> `models`, with Pydantic schemas at every boundary
- JWT-protected API with per-user data isolation, verified by tests
- Full audit trail of user actions, independent of the underlying data tables

23. Not Yet Implemented (Roadmap)

The following remain future work and are not part of the current build:

- Live market data feeds and real-time price updates (WebSockets)
- AI financial assistant, RAG document search (Ollama, ChromaDB)
- ML forecasting (expense/income/stock prediction)
- Dividend calendar and tax estimation
- PostgreSQL + Alembic migrations (currently SQLite with schema auto-create, dev-only)
- CI/CD pipeline, structured logging, error tracking, metrics/monitoring
- Two-factor authentication, refresh-token revocation, rate limiting
- Progressive Web App (offline mode, push notifications)